

Does our analysis match the client's data?

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Asahi India Glass gave us five files. We built a demand and forecasting system on top of them. This is the check that the numbers we show match the numbers they gave us.

The short answer

Yes, it matches.

Every quantity we show is identical to the client's own files. Units ordered, units despatched, units short, fill rate, the breakdown by branch, by glass type, by value class, by vehicle category, the month-by-month trend — all of it ties out exactly, not approximately.

There is one exception and it is small. The order value at one branch, Coimbatore, is ₹69,952 higher in our figures than in theirs. That is 0.0008% of the ₹857 crore total. The reason is that 44 of their order lines have no price at all, and we value those units while their file leaves them at zero. More on that below, but it is not an error in our arithmetic.

Getting to this point took three rounds of fixes. The check found genuine problems in how we were processing their data, and we corrected all three. That is worth knowing: the reconciliation was not a formality that passed first time.

How we checked

The obvious way to check a system is to read its code and convince yourself it looks right. We did not do that, because it proves very little.

Instead we went back to the raw files the client sent and recalculated everything from scratch, separately from the application, then compared the two sets of numbers. Open their order spreadsheet, add up the quantity column, and see whether it equals what our screen says. Where a chart groups demand by glass type, we rebuilt that grouping from their product master and compared category by category.

So there are two independent calculations of the same thing, and the question is simply whether they agree.

We also confirmed the five files themselves are untouched. Each one was fingerprinted and compared against the copy the client delivered, and all five are identical byte for byte. Nothing was cleaned, corrected or reformatted to make the numbers work.

The comparison covers the period from April 2025 to July 2026, which is 70 weeks and the whole span of their order book. Everything below is measured over that window.

The headline numbers

Across their whole network, all 53 branches and 2,063 products:

	Their files	Our analysis
Units ordered	2,602,392	2,602,392
Units despatched	2,133,961	2,133,961
Units short	504,052	504,052
Fill rate	82.00%	82.00%
Order value	₹857.10Cr	₹857.11Cr

These are not rounded to look neat. Two million six hundred and two thousand three hundred and ninety-two units on both sides.

The forecasting part of the system runs on a smaller slice, two branches and 136 products, and that reconciles too:

	Their files	Our analysis
Units ordered	146,814	146,814
Units despatched	129,559	129,559
Order value	₹59.95Cr	₹59.95Cr
Fill rate	88.25%	88.25%

On the smaller slice even the value matches exactly, because the 44 unpriced lines causing the network difference happen to sit outside it.

Chart by chart

A total matching is one thing. A manager is going to be shown charts, and a chart can be wrong in ways a total hides — the bars can be mislabelled, or one category can quietly swallow another, and the total still comes out right. So we checked every breakdown separately against the same raw files.

By branch. All 53 branches match exactly on units ordered and units despatched. On value, 52 of 53 match exactly. The one that doesn't is Coimbatore, and that's explained in the next section.

By glass type. Laminated 1,674,670 · Sidelite 577,327 · Backlite 347,748 · Tempered windscreen 6 · and 2,641 units under "Not in product master". All exact. That last bucket is new — see the fixes section.

By value class, by vehicle category, by top-10 SKU. Every line exact, and the ranking order is the same.

Over time. 16 months of demand, checked month by month. All 16 exact.

So the answer isn't just "the grand total is right". Every way we cut the data, it cuts the same way their file does.

The one number that differs

One figure on the whole system doesn't tie out, and it's worth being upfront about it rather than letting someone find it later.

Coimbatore's order value reads ₹190,690,964 in our analysis against ₹190,760,916 in the client's file. A difference of ₹69,952, which is 0.008% of the network's order value.

The cause is known precisely. There are 44 order lines at Coimbatore with no price on them — the rate column is blank. We cannot value a line with no rate, so those 44 lines contribute units but no money. The client's total carries a value for them from somewhere else in their system.

Two things are worth saying about this:

First, we could make the difference disappear in a minute by spreading an average price across those lines. We haven't, deliberately. The honest position is that 44 lines have no price in the data we were given, and the number should show that rather than hide it behind an estimate.

Second, those same 44 lines also have a blank despatch quantity. The client's own file is internally inconsistent here by ₹236,500 — more than three times our difference. So this is a gap in the source data, not a flaw in our arithmetic.

It also doesn't touch forecasting at all. Coimbatore isn't one of the two pilot branches, and every unit figure on the network is exact regardless.

Three things we found and fixed

This is the part I'd want a manager to see, because it says something about how the work was done. The reconciliation did not pass on the first attempt. Three real defects turned up, and all three are fixed.

Average price was computed the wrong way. We were averaging the price of each order line equally — a line for 2 units counted the same as a line for 2,000. That's not how an average price works. It's now weighted by quantity, which is what the client's own value column does. This moved value figures across the board.

A blank despatch was being read as a zero. When the despatch column is empty, that means nobody recorded it, not that nothing shipped. Treating it as zero inflated the shortfall and depressed the fill rate. Blanks are now counted as unknown and reported separately.

Products missing from the master were being dropped. Two SKUs that the client orders across 44 of their 53 branches don't appear in their own product master. Our charts were silently discarding those rows, so every product breakdown came up 2,641 units short of the headline total with nothing on screen explaining the gap. They now appear in a labelled bucket called "Not in product master" — visible, counted, and adding back to the total.

All three are covered by automated tests now, so they can't quietly come back.

Two false alarms

Worth mentioning, because both looked alarming for about ten minutes and neither was real.

The first branch-level comparison showed roughly 60% of branches mismatching on units. The cause was my own check — I'd forgotten the date filter, so I was comparing their filtered window against our full history. With the filter applied, all 53 matched.

The second was the monthly trend, where all 16 months looked wrong. Again my check, not the system. The data is held in weeks, and a week that straddles two months has to be assigned to one of them. The standard rule is that the week belongs to whichever month contains its Thursday. I'd compared against plain calendar months. Applying the correct rule, all 16 months matched exactly.

I'm including these because the distinction matters: a number disagreeing doesn't automatically mean the system is wrong. Both of these were the test being wrong, and both were run again properly.

How accurate is the forecast?

Separate question from whether the data matches, and worth keeping separate. The data reconciles exactly. The forecast is a prediction, and predictions have error.

Measured on the pilot scope, 261 branch-and-product lines:

Horizon	Accuracy
1 month out	71.45%
3 months out	82.95%
6 months out	89.55%

The six-month figure corresponds to an error rate of 10.45%. 159 of the 261 lines are at or above the 85% target. 249 lines get better as the horizon lengthens, which is expected — monthly noise averages out over six months.

Two honest caveats. 40 lines out of 272 are beaten by a simple baseline, meaning for those specific series the sophisticated model isn't earning its keep yet. And every accuracy figure here was measured on held-out historical data using a proper time-ordered split, never a random one — a random split would let the model see the future and would produce a flattering number that falls apart in production.

Where this leaves us

The answer to your manager's question is yes. Totals, every branch, every chart, every month — all reconcile to the client's own files. One value figure differs by 0.008% and we can explain exactly why, down to the 44 order lines causing it.

Two things worth asking the client:

1. The 44 Coimbatore lines with no price and no despatch quantity — is there a source for those, or should they be treated as incomplete?
2. The two SKUs ordered across 44 branches but absent from their product master — should they be added, or are they intentionally outside it?

Neither blocks anything. Both are gaps in their data that our reconciliation surfaced, which is arguably useful output in its own right.